

Vortex Gaussians: Real-Time Simulation-Driven Fire and Smoke as Native Gaussian-Splatting Content

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Abstract

Fire and smoke remain conspicuously absent from Gaussian-splatting (GS) scenes: existing physics-integrated GS methods target solids and liquids, GS fire methods reconstruct flames from video rather than simulating them, and the one system that couples forward combustion simulation with GS scenes composites ray-marched volumes over an opaque backdrop at seconds per frame. We present the first system that generates fire and smoke by forward physical simulation in real time and renders them as *native* Gaussian-splatting content: the fire’s Gaussians share one global depth order and one emission–absorption blend state with a Gaussian-splat environment, so a foreground scene object correctly occludes the flame and the flame occludes the scene behind it. Our key insight is a structural isomorphism: Lagrangian vortex particles and 3D Gaussians are the same class of point primitive. We therefore let one particle set serve simultaneously as simulation state and render primitive—position maps to the Gaussian mean, temperature to emissive blackbody color, density to opacity, and local velocity gradient to an anisotropic covariance evolved in closed form—eliminating voxelization, meshing, and any simulation-to-renderer conversion. Dynamics come from a two-level layered vortex solver inspired by production fire pipelines but recast entirely in vorticity–streamfunction form with no pressure variable: each of N view-aligned slices runs a 2D vortex-in-cell simulation with a baroclinic temperature–buoyancy–vorticity loop at $O(N_p + G)$ cost, a coarse global solver supplies bulk coherence, and—new to layered fire—a bidirectional inter-slice vorticity-coupling operator reintroduces the out-of-plane transport that stacking drops, measurably raising cross-slice coherence. All flame, smoke, ember, and scene Gaussians share one depth-sorted emission–absorption rasterizer with HDR blackbody shading; loaded reconstructions are merged into the single global depth order by an asymmetric exact/bucketed two-stream interleave at no measured overhead. Our single-file WebGL2 prototype sustains real-time rates on a consumer laptop, supports interactive in-scene ignition and control of wind and coupling, and we quantify slice orientation and inter-slice coherence with headless probes and ablate the major design choices.

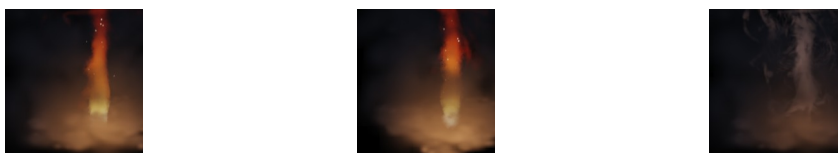


Figure 1. Real-time, simulation-driven fire and smoke rendered as native Gaussian-splatting content. Left: a flame driven by nine layered 2D vortex-in-cell simulations, in which every vortex particle is an emissive anisotropic Gaussian (white-hot core $\rightarrow$ orange $\rightarrow$ dark-red tips, with embers). Middle: the same fire from an orbiting camera; view-aligned slices and per-primitive depth sorting give correct parallax for the rendering camera against a dark 3DGS background scene that the fire locally illuminates. Right: the identical particle set re-shaded as smoke (absorption-dominated compositing with warm underlighting). All results run in real time in a browser. *[TODO: Replace with final high-quality captures; add accompanying video.]*

1 Introduction

Radiance-field capture has made it routine to bring real environments into the computer as 3D Gaussian-splatting (3DGS) scenes that render photorealistically in real time (Kerbl et al. 2023). A growing body of work now asks how such scenes can be brought to *life*: physics-integrated Gaussian methods animate elastic solids and liquids by attaching continuum solvers to the reconstructed primitives (Xie et al. 2024; Feng et al. 2025; Zhong et al. 2024; Jiang et al. 2024). Yet the most visually iconic dynamic phenomena—fire and smoke—remain conspicuously missing from this ecosystem. Existing Gaussian representations of flames are *inverse*: they reconstruct an observed fire from video (Nazarenius et al. 2025; Shui et al. 2024) and can only replay it, as do neural volumetric methods for smoke and fluids (Chu et al. 2022; Yu et al. 2023; Gao et al. 2025). The one system that synthesizes new fire in reconstructed GS scenes, FieryGS (Shen et al. 2026), runs an offline 256^3 Eulerian solver and ray-marches the resulting volume over the GS scene treated as an opaque backdrop, at seconds per frame—two orders of magnitude from interactive.

This gap is not accidental. Gaseous combustion has neither a surface to mesh nor persistent material points to track, so the standard physics-to-GS recipes (bind Gaussians to a deformation field; skin them to simulation particles) do not directly apply, and volumetric emission seems to demand ray marching.

Our starting point is an observation about *representation*: a Lagrangian vortex particle and a 3D Gaussian are the same kind of object. Both are unordered point primitives carrying a position and a handful of per-point attributes; both live in flat GPU buffers; and both are consumed by a sort-and-blend operator (Biot–Savart summation over particles; front-to-back alpha compositing over splats). We exploit this isomorphism literally: **one particle set is simultaneously the simulation state and the render primitive**. Each vortex particle carries circulation Γ , temperature T , and density d ; the renderer reads the very same particle as an emissive anisotropic Gaussian whose mean is the particle position, whose color is a blackbody function of T , whose opacity follows d , and whose covariance is evolved by the local velocity gradient. No voxelization, no marching cubes, no simulation-to-renderer resampling of any kind separates physics from pixels.

For dynamics we revisit a production idea. Industrial Light & Magic’s film fire system (Horvath and Geiger 2009) showed that a stack of independent, camera-facing 2D slice simulations, driven by a coarse directable 3D pass, is perceptually sufficient for even hero-shot fire—but it ran on a GPU farm at ~ 20 s per slice per frame. We make this two-level architecture real-time and fully Lagrangian: each slice runs a 2D *vortex-in-cell* (VIC) solver in which buoyancy injects circulation into particles through the baroclinic term $\beta \partial T / \partial x$, a small streamfunction Poisson solve reconstructs a divergence-free velocity at $O(N_p + G)$ cost (replacing $O(N_p^2)$ Biot–Savart summation), and a coarse global solver nudges all slices toward one coherent bulk motion. Because the slices’ particles are already Gaussians, the entire stack—flame, smoke, embers, and the static background scene—feeds a single depth-sorted emission–absorption rasterizer with HDR blackbody shading, so the fire correctly occludes, is occluded by, and locally illuminates the environment.

We validate the system with an interactive prototype implemented as a single-file WebGL2 application whose simulation runs in JavaScript on the CPU, demonstrating that the method is cheap enough for real time even without GPU compute. We further contribute a decoupled per-layer interface that hot-swaps each slice between the physical solver and a procedural curl-noise source at runtime, which we use both as a development methodology and as an ablation instrument.

In summary, our contributions are:

- Extending the particle–Gaussian unification to media without material points.** Physics-integrated GS established that simulation particles can double as rendered Gaussians for solids and liquids (Xie et al. 2024; Feng et al. 2025); gases have neither a rest configuration nor persistent material points, which is precisely what *vorticity-carrying* particles supply. We build, to our knowledge, the first system in which forward-simulated, emissive gaseous phenomena exist as GS-native content, and we demonstrate the pattern across three instances— flame, smoke, and embers—driven through one particle contract and one rasterizer: one attribute mapping $(x, T, d, u, \nabla u) \mapsto (\mu, c, \alpha, \Sigma)$ with zero conversion passes, where *GS-native* means concretely the same instance format, the same per-primitive depth sort, and the same front-to-back emission–absorption blend as the 3DGS forward pass (§3, §5).
- A pressure-free two-level layered vortex-in-cell solver.** Both scales—the coarse global solve and every refinement slice—are uniformly in vorticity–streamfunction form and driven end to end by the single baroclinic source, with *no pressure variable anywhere*; this is the precise sense in which our method is not “ILM at low resolution,” since the coarse PIC/FLIP pass and grid-Navier–Stokes slices of Horvath and Geiger (2009) are both pressure-projected with vorticity entering only as a confinement correction. A continuously tunable velocity nudge to the coarse field supplies global coherence, and the coarse-grid vortex particles carry Lagrangian state that survives slice re-orientation without re-simulation (§4).
- A real-time inter-slice vorticity-coupling operator.** We introduce a bidirectional, circulation-conserving coupling that reconstructs the out-of-plane second derivative $\partial^2 \omega / \partial z^2$ a stacked-2D model structurally drops, by a z-Laplacian exchange of vorticity across adjacent view-aligned slices. Production layered fire (Horvath and Geiger 2009) has *no* slice-to-slice communication (coherence there comes only from a one-way coarse broadcast); ours is, to our knowledge, the first inter-slice vorticity exchange for layered emissive fire, and it demonstrably raises a cross-slice velocity-coherence probe monotonically with its strength (§4.2, §7). It partially compensates for—but does not restore—true 3D vortex stretching, which remains absent (§8).
- Gaussian-native volumetric fire shading.** Flow-driven anisotropy: adapting covariance-from-deformation (Xie et al. 2024) to rest-configuration-free media by evolving $\dot{\Sigma} = L \Sigma + \Sigma L^T$ incrementally, with renormalization and clamps acting on the

rendered quantity itself, plus an optional trajectory-filament stretch—combined with blackbody HDR emission–absorption compositing, bloom, and filmic tone mapping (§5).

- **Unified composition with GS scenes via an asymmetric two-stream depth merge.** Fire, smoke, embers, and environment Gaussians share one global depth order and one blend state: the small dynamic set is sorted exactly while a loaded reconstruction is counting-sorted into K view-depth buckets in $O(m + K)$, and the two pre-ordered streams are interleaved near-to-far—precision matched to where ordering error is visible. This yields per-primitive mutual occlusion at no measured overhead over a naive draw-after pass, and enables interactive *in-scene* ignition: a fire lit on a reconstructed surface is occluded per primitive by the geometry it sits inside (§5.5); prior art composites fire *over* GS scenes as an opaque backdrop (Shen et al. 2026).
- **An evaluation of the layered-Gaussian design space.** A quantitative viewpoint-isotropy probe for slice orientation (view-aligned vs. world-fixed), the inter-slice coherence probe above, and qualitative ablations over coupling strength, covariance dynamics, and a matched-budget kinematic 3D alternative (§7).

2 Related Work

2.1 Fire and Smoke Simulation

Visually plausible gaseous simulation in graphics descends from the unconditionally stable semi-Lagrangian advection of Stam (1999), extended with vorticity confinement to fight numerical dissipation (Fedkiw et al. 2001; Steinhoff and Underhill 1994) and specialized to combustion by Nguyen et al. (2002), whose separate fuel/hot-product phases and blackbody radiance remain the canonical physically based fire model. Real-time grid solvers moved to the GPU early (Harris 2004; Crane et al. 2007).

Vortex methods discretize the vorticity form of the Euler equations on Lagrangian elements—particles (Chorin 1973; Cottet and Koumoutsakos 2000; Park and Kim 2005; Selle et al. 2005), filaments (Angelidis and Neyret 2005; Weißmann and Pinkall 2010), or sheets (Pfaff et al. 2012; Brochu et al. 2012)—which auto-satisfy incompressibility in reconstruction, need no pressure projection, and preserve turbulent detail with striking economy. Their classical bottleneck is velocity recovery: naive Biot–Savart summation is $O(N_p^2)$, mitigated by treecodes and fast multipole methods (Brochu et al. 2012) or by particle–grid hybrids (vortex-in-cell) (Cottet and Koumoutsakos 2000). Vorticity-based solvers continue to advance, most recently through flow-map formulations (Wang et al. 2024). We adopt a vortex-in-cell formulation per slice, obtaining $O(N_p + G)$ velocity reconstruction on grids small enough for JavaScript, and we inherit the Lagrangian virtue that matters most here: *the solver’s degrees of freedom are already renderable points*. Brochu et al. (2012) argued that simulation cost should scale with rendered detail and named real-time vortex-based fire an open problem; our system is one answer.

Production layered fire. Horvath and Geiger (2009) decompose film fire into a coarse, directable 3D particle pass (a PIC/FLIP variant (Zhu and Bridson 2005)) and a stack of independent,

camera-facing 2D Navier–Stokes slice simulations (typically $\sim 2048^2$; film examples at 2048×1556 with 32–128 slices), composited near-to-far with Planck blackbody shading. The insight that fine detail perpendicular to the view plane is perceptually invisible justifies 2D refinement. Their slices never exchange simulation state with one another; coherence comes from the shared coarse forcing, splatted with a Gaussian depth kernel spanning ~ 8 slices. The system is offline (~ 20 s per slice per frame on a GPU farm), renders slices as textures composited against holdout mattes, and its stated limitations—fast-moving cameras, flows perpendicular to the view plane, no inter-slice communication—map directly onto the design decisions we revisit. We replace per-slice grid Navier–Stokes with 2D vortex particles whose Lagrangian state survives slice re-orientation, replace slice textures with Gaussians shared with the environment, and replace the fixed projection kernel with a coarse global *solver* whose influence is a continuously tunable velocity blend.

Layered and billboard volume rendering. Half-angle slicing, volumetric billboards, and spherical billboards render volumes as stacks of blended proxies (Decaudin and Neyret 2009; Umenhoffer et al. 2006); games have long shipped 2.5D flipbook and card-based fire. These techniques require the three ingredients our renderer gets for free from splatting: depth-sorted blending, parallax from depth offsets, and camera-facing orientation.

2.2 Gaussian Splatting

3DGS (Kerbl et al. 2023) represents radiance fields as anisotropic Gaussians rasterized by tile-based, depth-sorted front-to-back alpha blending, following the EWA volume-splatting lineage (Zwicker et al. 2001); anti-aliased variants (Yu et al. 2024) and physically based volumetric-primitive transport (Condor et al. 2025) refine the formulation. Condor et al. (2025) in particular show Gaussian mixtures are a sound basis for *emissive* media under ray tracing; we target the rasterized, real-time end of that trade-off. Dynamic-scene extensions obtain motion from data—per-frame optimization (Luiten et al. 2024) or learned deformation fields (Wu et al. 2024)—and thus replay observed dynamics rather than generate new ones.

Physics-integrated Gaussians animate reconstructed scenes forward in time: PhysGaussian evolves Gaussian kernels under MPM continuum mechanics (Xie et al. 2024), Gaussian Splashing couples position-based fluids with splat rendering (Feng et al. 2025), Spring-Gaus embeds spring-mass systems (Zhong et al. 2024), and VR-GS makes such coupling interactive (Jiang et al. 2024). All target solids or liquids—media with material points or surfaces—and none address optically thin emissive volumes. Orthogonally, Gaussian Fluids (Xing et al. 2025) uses Gaussians as *basis functions* for the velocity field itself, solving incompressible flow by per-step optimization at 38–228 s per step, without any rendering; we use Gaussians as *render primitives* carried by a classical real-time solver. The two works bracket our claim from opposite sides: Gaussians suit both fluid simulation state and fluid appearance, and our contribution is to make one particle set serve both at once.

Gaussians for fire. FlameGS (Shui et al. 2024) and Gaussians on Fire (Nazarenius et al. 2025) reconstruct emissive flame light fields from images; PINF (Chu et al. 2022), HyFluid (Yu et al. 2023), FluidNexus (Gao et al. 2025), and Vid2Fluid (Zhao et al. 2025) recover smoke or fluid dynamics from video with physics priors. These are inverse problems. FieryGS (Shen et al. 2026) is the closest prior system and the only one that *synthesizes* fire in GS scenes: an MLLM assigns combustion properties to a reconstructed scene, a 256^3 Eulerian solver simulates fire, and

a ray-marched volume is composited over the GS scene attenuated as an opaque background, at 2.37 s per frame (0.84 s of which is the fire/smoke ray march). Our system differs on every axis (Table 1): Lagrangian rather than Eulerian simulation, per-primitive depth interleaving rather than backdrop compositing, rasterization rather than ray marching, and real-time rather than offline. FieryGS’s orthogonal capabilities—material inference, charring—are complementary to our pipeline.

Table 1. Positioning. “GS-native” means the phenomenon is rendered as Gaussian primitives depth-interleaved with the scene, not composited over it. **[TODO: verify FieryGS numbers against camera-ready version.]**

	gaseous emissive	forward physics	real time	GS- native	scene composited
Horvath and Geiger (2009)	✓	✓	—	—	matte
PhysGauss ian (Xie et al. 2024)	—	✓	—	✓	✓
G. Splashi ng (Feng et al. 2025)	—	✓	—	✓	✓
G. Fluids (Xing et al. 2025)	—	✓	—	—	—
Fire recon. (Na zarenus et al. 2025)	✓	—	(render)	✓	—
FieryGS (S hen et al. 2026)	✓	✓	—	—	backdrop
Ours	✓	✓	✓	✓	✓

3 Overview

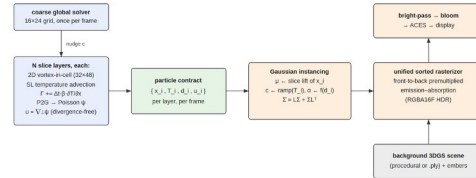


Figure 2. System overview. A coarse global vortex solver drives N per-slice 2D vortex-in-cell simulations (left, §4). Each slice exposes its particles through a minimal contract (center); the

renderer lifts each particle directly into an emissive anisotropic Gaussian (§5) and rasterizes all Gaussians—fire, smoke, embers, and the background 3DGS scene—in one depth-sorted emission–absorption pass with HDR post-processing. There is no voxelization or resampling anywhere in the pipeline.

Figure 2 summarizes the system. The domain is a stack of N parallel slice layers (default $N=9$) spanning the fire volume. Once per frame we advance a coarse global solver (§4.2) and then each layer’s 2D vortex-in-cell simulation (§4.1); every layer exposes its particles through the contract

$$\text{Layer}.\text{step}(\Delta t) \rightarrow \{x_i \in \mathbb{R}^2, T_i, d_i, u_i \in \mathbb{R}^2\},$$

i.e., in-plane position, temperature, density, and velocity. The renderer is the *only* consumer of this contract: it lifts each particle into world space using the layer’s slice frame (§4.3) and instances one emissive Gaussian per particle (§5). Because the contract is source-agnostic, any layer can be hot-swapped at runtime between the physical solver and a cheap procedural curl-noise plume (Bridson et al. 2007)—a property we exploit for incremental system verification and for ablations (§7).

4 Two-Level Layered Vortex Simulation

4.1 Per-Slice Vortex-in-Cell Dynamics

Each slice simulates buoyant flow in the vorticity–streamfunction form of the incompressible Euler equations. In 2D, vorticity is the scalar $\omega = \partial_x u_y - \partial_y u_x$ and the vortex-stretching term vanishes identically, leaving

$$\frac{\partial \omega}{\partial t} + (u \cdot \nabla) \omega = (\nabla \times f_b)_z = \beta \frac{\partial T}{\partial x}, \quad (1)$$

where the right-hand side is the *baroclinic* source obtained by taking the curl of Boussinesq buoyancy $f_b = \beta(T - T_{amb})\hat{y}$: horizontal temperature gradients at the plume edge inject the counter-rotating vortex pairs that roll rising columns into mushroom caps. This single term closes the temperature $\rightarrow$ buoyancy $\rightarrow$ vorticity loop that gives fire its characteristic motion (Nguyen et al. 2002; Selle et al. 2005).

Vorticity is carried by particles as circulation Γ_i . Per frame, each slice performs six steps on an $n_x \times n_y = 32 \times 48$ grid over a 2×3 world-unit domain:

1. *Temperature advection.* The grid temperature field T is advected semi-Lagrangianly (Stam 1999) with the previous frame’s velocity, heated near the source by $T += \Delta t s_0 \exp(-\kappa_s \tilde{x}^2)$ in the source band, cooled by $T^* = (1 - 1/2\kappa_c \Delta t)$, and clamped.
2. *Baroclinic injection.* Each particle accumulates circulation from the local horizontal temperature gradient (centered differences of the advected field):

$$\Gamma_i \leftarrow (1 - \lambda_r \Delta t) \Gamma_i + \Delta t \beta \frac{\partial T}{\partial x} \Big|_{x_i}. \quad (2)$$

3. *P2G*. Particle circulations are deposited onto the grid vorticity ω with bilinear weights.
4. *Streamfunction solve*. We solve the Poisson equation $\nabla^2 \psi = -\omega$ with $K = 26$ Jacobi iterations and homogeneous Dirichlet boundaries.
5. *Velocity reconstruction*. $u = \nabla^\perp \psi = (\partial_y \psi, -\partial_x \psi)$ by centered differences; the result is divergence-free by construction for *any* ψ , so the truncated Jacobi solve trades only vorticity accuracy, never incompressibility.
6. *G2P and advection*. Particles gather the grid velocity bilinearly, add direct buoyancy $\beta_b T_i \hat{y}$, a small divergence-free curl-noise perturbation (Bridson et al. 2007), wind, and the global nudge of §4.2, then advect with explicit Euler.

Note that buoyancy appears twice by design, with independent gains: the baroclinic term (Eq. 2, gain β) generates the *rotational* response—edge rolling and mushroom caps—while the direct term (gain β_b) is an additional non-solenoidal control force for bulk rise. Decoupling the two is an artistic choice, not a derivation: in pure vorticity form the body force would enter only through its curl.

Steps 3–5 constitute the vortex-in-cell velocity reconstruction (Cottet and Koumoutsakos 2000): they replace the $O(N_p^2)$ regularized Biot–Savart sum $u(x) = \sum_j \Gamma_j K_\varepsilon(x - x_j)$ (vortex-blob kernel $K_\varepsilon(r) = 1/2\pi(-r_y, r_x)/\|r\|^2 + \varepsilon^2$ (Chorin 1973; Beale and Majda 1985)) with an $O(N_p + G)$ transfer–solve–gather cycle, where G is the (small) grid size. At 600 particles and a 32×48 grid per slice, nine slices leave ample headroom within a frame budget even in single-threaded JavaScript, whereas the naive Biot–Savart summation (retained in our codebase as a reference path) scales quadratically in the particle count ([**TODO: exact timing table on named hardware, incl. VIC vs. Biot–Savart at matched particle counts**]).

Thermodynamics and lifecycle. Particles carry their own temperature, cooled nonlinearly, $dT_i/dt = -\kappa T_i^2$, which concentrates brightness at the hot base and produces the natural fast fade toward the flame tips; density decays exponentially. Particles are recycled when cold, exhausted, or out of bounds, and re-emitted at the source (~ 6 particles per layer per frame up to a 600-particle cap), so the particle count—and hence the Gaussian count—stays bounded.

4.2 Coarse Global Solver and Inter-Layer Coupling

Independent slices produce N *uncorrelated* fires. Production layered fire solved this with a splat kernel spanning neighboring slices (Horvath and Geiger 2009); we instead couple on the simulation side. A single coarse solver (grid 16×24 , same structure as §4.1 but grid-only, with a quasi-static vorticity estimate $\omega = \beta \partial T / \partial x$ and $K = 30$ Jacobi iterations) is advanced once per frame to produce a global velocity field U representing the directable bulk motion of the whole fire. During each slice’s G2P step, every particle is nudged toward the global field:

$$u_i \leftarrow u_i + c(U(x_i) - u_i), c \in [0, 1]. \quad (3)$$

$c=0$ leaves slices fully independent (each layer visibly “its own fire”); $c=1$ slaves all slices to the coarse solve (one coherent but overly smooth column); the default $c=0.5$ retains per-slice turbulent detail on top of a shared bulk motion (Figure 6). This is the two-level decomposition of Horvath and Geiger (2009)—coarse directable motion plus per-slice refinement—with the coupling expressed as a first-class, continuously tunable simulation parameter rather than a fixed projection kernel. The coarse field also offers a single low-dimensional handle to which direction forces *can* be confined; our prototype currently applies wind and source motion per-particle as well.

Inter-slice vorticity coupling.

Equation 3 couples every slice to a shared *global* field, but it does not couple slices to their immediate *neighbors*: two adjacent slices can follow the same bulk motion while their turbulent detail stays uncorrelated, and stacked-2D dynamics structurally drop the out-of-plane transport $\partial^2 \omega / \partial z^2$ that would tie them together. We reintroduce this dropped coupling directly. After each slice deposits its particle circulation onto the grid vorticity ω_k (step 3), we add a z-Laplacian exchange with the adjacent slices’ vorticity from the previous frame,

$$\omega_k \leftarrow \omega_k + \Delta t \kappa_z (\omega_{k-1}^{t-1} + \omega_{k+1}^{t-1} - 2\omega_k^{t-1}), (4)$$

with zero-flux (Neumann) ends so that total circulation $\sum_k \omega_k$ is conserved. This is a symmetric, one-frame-lagged reconstruction of the out-of-plane second derivative the model drops; it makes concentrated vortices span several slices, so the stack evolves as a single 3D volume rather than N independent sheets. Production layered fire (Horvath and Geiger 2009) explicitly has no such slice-to-slice communication—its slices are coupled only through the one-way coarse broadcast—so Eq. 4 is, to our knowledge, the first inter-slice vorticity exchange for layered emissive fire. The two mechanisms are complementary and orthogonal: Eq. 3 imposes global bulk coherence, Eq. 4 imposes local neighbor coherence. We report a cross-slice velocity-coherence probe as a function of κ_z in §7. Crucially, Eq. 4 models the *coupling* that stacking drops; it does *not* recover true 3D vortex stretching $(\omega \cdot \nabla)u$, whose spectral consequences we treat honestly in §8.

4.3 Slice Geometry

Layer $k \in \{0, \dots, N-1\}$ is assigned the depth offset $z_k = (k / (N-1) - 1/2) D$ along the stack axis, with a mild per-layer opacity fade $1 - 0.3|z_k|$. We support two slice frames:

View-aligned (default).

The in-plane axes are the camera’s horizontal right vector and the world up vector, and layers are offset along the horizontal camera-forward direction. Slices re-orient continuously as the camera orbits, so the stack always presents its face—the choice of Horvath and Geiger (2009), without their per-view re-simulation, because our simulation state lives on particles rather than on slice rasters.

World-fixed.

Slices are pinned to a world plane and offset along the world normal. Orbiting reveals the stack edge-on and the volume collapses into stripes (Figure 5).

In a headless probe of peak image brightness—the maximum per-pixel summed RGB value (range 0–765) read back from the tone-mapped framebuffer—the view-aligned frame is isotropic across viewpoints (side/front = 758 / 758) while the world-fixed frame degrades to 531 from the side (§7). Because re-orientation only changes the *lift* of in-plane particle coordinates into world space—not the simulation state—view-aligned slicing costs nothing and introduces no re-simulation popping; the known failure modes of view-locked slices under fast camera motion (Horvath and Geiger 2009) are correspondingly milder but not eliminated (§8).

5 Emissive Gaussian Rendering

5.1 Particle-to-Gaussian Mapping

Table 2. *The complete simulation-to-rendering mapping. Every attribute of the rendered Gaussian is read directly off the simulation particle; there is no intermediate representation.*

particle attribute	Gaussian attribute	mechanism
position x_i + layer k	mean μ_i	slice-frame lift (§4.3)
temperature T_i	color c_i	blackbody ramp (§5.4)
density d_i	opacity α_i	§5.3
velocity gradient L_i	covariance Σ_i	Eq. 5
trajectory history	filament axis	Eq. 6

Table 2 lists the mapping. Each particle is instanced as a camera-facing quad whose vertex shader expands the unit square along the Gaussian’s major and minor semi-axes and projects with the full perspective camera; the fragment shader evaluates the Gaussian footprint $g = \exp(-3r^2)$ modulated by a four-octave value-noise FBM detail factor that breaks the characteristic “blobby” look of isolated splats. Instance data is a flat 14-float record (position, two axes, fade, T , d , seed, type)—the direct render-side realization of the simulation contract.

5.2 Flow-Driven Covariance

Isotropic splats read as bubbles; real flames are sheared into streaks and wisps by the flow. Analogously to how PhysGaussian transforms covariances by the deformation gradient (Xie et al. 2024), we evolve each particle’s in-plane 2×2 covariance directly by the local velocity gradient $L = \nabla u$ (measured from the slice grid by centered differences):

$$\dot{\Sigma} = L \Sigma + \Sigma L^T, (5)$$

integrated explicitly with gain k_Σ , relaxed toward identity at rate k_r , and stabilized in closed form: after a 2×2 eigen-decomposition we renormalize $\det \Sigma = 1$ and clamp the leading eigenvalue $\lambda_0 \leq \kappa_{max}$ (with $\lambda_1 = 1 / \lambda_0$ from the determinant constraint, this bounds the splat’s semi-

axis ratio a/b at κ_{max}). Under $tr L=0$ exact integration would preserve $det \Sigma$; the renormalization removes explicit-integration drift. Equation 5 is the differential form of PhysGaussian’s $\Sigma = F \Sigma_0 F^T$ (Xie et al. 2024)—but for a gas, F and Σ_0 are ill-defined absent a rest configuration, which makes the incremental form the natural one; and because the clamps act on the rendered quantity itself, the unbounded-elongation (“sliver”) failure mode reported for Gaussians under sustained shear (Xie et al. 2024; Feng et al. 2025) cannot occur. The splat’s semi-axes are then $a = s\sqrt{\lambda_0}$, $b = s\sqrt{\lambda_1}$ along the eigenvectors, so vortical shear visibly stretches blobs into licks of flame (Figure 7). Attaching per-particle anisotropy to a coarse solve echoes anisotropic turbulence particles (Pfaff et al. 2010), with the anisotropy here doubling as the render primitive’s footprint.

As an optional third tier, a trajectory *filament* stretch elongates each Gaussian along its recent path using an exponentially lagged shadow position p^{lag} :

$$p^{lag} \leftarrow p^{lag} + \eta(p - p^{lag}), a = s + k_f \|p - p^{lag}\|, (6)$$

turning the particle set into short path-integrated streaks reminiscent of vortex filaments (Angelidis and Neyret 2005; Weißmann and Pinkall 2010)—an instantaneous-shear complement particularly effective for embers and fast plumes.

5.3 Unified Sorted Compositing

All dynamic and procedural-scene Gaussians—flame, smoke, embers, and the 640-splat background environment—are gathered into one buffer, sorted front-to-back by view depth, and rasterized in a single instanced draw into an RGBA16F target cleared to zero coverage. Fragments output premultiplied $(c_i \alpha_i, \alpha_i)$ and are blended with

$$C \leftarrow C + (1 - A)c_i \alpha_i, A \leftarrow A + (1 - A)\alpha_i, (7)$$

(GL blend ONE_MINUS_DST_ALPHA, ONE)—exactly the front-to-back emission-absorption compositing of the 3DGS forward pass (Kerbl et al. 2023; Zwicker et al. 2001). The two media differ only in their opacity models: fire is optically thin and emission-dominated, so its Gaussians carry low α ($\alpha_i = clamp((\alpha_0 + \alpha_1 d_i) T_i D_i, 0, \alpha_{max})$ with small α_{max}), accumulating radiance while barely occluding; smoke is absorption-dominated with high α and a density-modulated gray albedo warmed by residual temperature. Since emission and absorption are independent per-Gaussian channels, the *same* particle state spans the fire-to-smoke range under two scalar knobs (emission gain, absorption albedo) with no change to geometry, sort, or solver—consistent with Kirchhoff’s law, hot particles emit and cooled ones absorb—so a single plume can transition from emissive flame at its hot base to absorptive smoke at its cooled tips. Because occlusion is resolved per primitive by the shared sort, the fire correctly sits *inside* the scene: environment splats in front of the flame occlude it, and the flame occludes those behind (Figure 1, middle)—in contrast to compositing a ray-marched volume over the scene as a whole (Shen et al. 2026). Procedural environment splats share the sorted buffer directly; large *loaded* reconstructions are merged into the same global depth order by bucketed interleaving (§5.5, Figure 4).

5.4 Blackbody HDR Shading and Post-Processing

Fire emission follows incandescence: we map particle temperature through a hand-tuned five-stop ramp approximating Planck blackbody colors (Nguyen et al. 2002; Pegoraro and Parker 2006; Horvath and Geiger 2009) (dark ember red → red → orange → yellow → near-white), scaled by an HDR emissive gain and a low-frequency flicker. Radiance accumulates far beyond display range, so the composited scene passes through a standard HDR chain: bright-pass extraction, five iterations of half-resolution separable Gaussian blur (bloom), then ACES filmic tone mapping (Narkowicz 2016), gamma, vignette, and a 1/255 dither. Combined with the nonlinear particle cooling of §4.1, this reproduces the canonical luminance structure of real flames—saturated white-hot core, graded orange body, dark red tips—without any texture assets (Figure 9).

5.5 Scene Composition and Secondary Effects

The environment is itself Gaussian content: either a procedurally scattered dark scene (640 flattened ground and enclosure splats) or a reconstructed 3DGS capture loaded from a standard Inria .ply (scales exponentiated, opacities sigmoided, DC spherical-harmonic color; subsampled to a budget of up to 120k splats). Procedural environment splats join the sorted buffer of §5.3 directly, giving per-primitive mutual occlusion with the fire. Loaded reconstructions are merged into the same global depth order by *depth bucketing*: each frame, scene splats are counting-sorted into $K = 96$ view-depth buckets in $O(n + K)$, and the renderer walks the buckets near-to-far, interleaving ranges of the (exactly sorted) dynamic stream with ranges of the bucketed scene stream under the shared blend state, coalescing adjacent same-stream ranges to minimize state changes. Order is thus exact within the dynamic stream and bucket-granular between and within the streams—strictly finer than the per-slice compositing granularity of layered production fire (Horvath and Geiger 2009). The bucketing adds no measurable cost: with the interleave on vs. a naive draw-after pass we measure 3.5 vs. 3.6 ms per frame at 3.2k scene splats and 13.0 vs. 13.3 ms at 79k splats (informal, development laptop; Figure 4). **[TODO: repeat the occlusion figure with a real captured .ply reconstruction in place of the synthetic test scene.]**

In-scene ignition. Because the fire shares the scene’s depth order rather than sitting in front of a backdrop, a source can be lit *on a reconstructed surface*: the plume then rises inside the captured geometry and is occluded per primitive by the objects around it (Figure 4, left). This in-place authoring is structurally unavailable to systems that composite fire over an opaque scene (Shen et al. 2026). A single source is shown here; multiple concurrent sources sharing the coarse field are a small extension (**[TODO: demonstrate multi-source in-scene ignition on a real capture; note how concurrent sources partition the 16×24 coarse field]**).

The fire acts as a light source on the scene through an inexpensive screen-space model: environment splats are warmed by a radial falloff around the flame’s screen position with a floor-proximity boost—a deliberately cheap stand-in for volumetric light transport that

nevertheless anchors the flame in the scene (§8). Sparse ember particles (velocity-aligned, twinkling Gaussians in the shared sorted pass), plus two screen-space effects—a mirrored planar ground reflection and an FBM heat-haze refraction above the flame—complete the presentation; all are optional.

6 Implementation

The entire system—simulation, sorting, rendering, UI—is a single dependency-free HTML file: simulation and per-frame depth sort run in JavaScript on the CPU; rendering is WebGL2 with instanced quads, an RGBA16F HDR target (8-bit fallback), and hand-written camera matrices. We emphasize what this implies: the method needs no GPU compute, no CUDA, and no preprocessing, and still reaches real-time rates (§7). Default configuration: $N=9$ layers $\times$ 600 particles (~ 5.4 k flame Gaussians), 640 scene splats, ≤ 160 embers—about 6.2k sorted Gaussians per frame, plus up to 120k additional splats when a reconstructed scene is loaded (merged into the global depth order by $O(n)$ bucketing; §5.5). Table 3 lists the main parameters. A headless harness drives the demo without a display and reads back probe images for the quantitative measurements in §7.

Table 3. Key parameters (defaults). World domain 2×3 units.

symbol	meaning	default	section
N	slice layers	9	§4.3
N_p	particles per layer	600	§4.1
$n_x \times n_y$	slice grid	32×48	§4.1
—	coarse grid	16×24	§4.2
K	Jacobi iterations	26 / 30	§4.1
β	baroclinic gain	1.4	Eq. 2
β_b	direct buoyancy	0.9	§4.1
κ	cooling rate	1.2	§4.1
c	global (coarse) coupling	0.5	Eq. 3
κ_z	inter-slice coupling	0 (opt.)	Eq. 4
D	stack depth	0.7	§4.3
k_Σ	covariance gain	1.3	Eq. 5
κ_{max}	eigenvalue clamp (bounds axis ratio)	3.6	Eq. 5
—	emissive gain / bloom / threshold	1.4 / 1.0 / 0.95	§5.4

7 Results and Evaluation

Performance.

At default settings the prototype runs at display rate in informal use on a consumer laptop, with the simulation single-threaded in JavaScript; we expect the per-frame cost to be dominated by the nine 32×48 Poisson solves and the CPU depth sort of $\sim 6.2k$ instances, but systematic measurements are pending. **[TODO: Benchmark table on named hardware**

(laptop iGPU + desktop dGPU): FPS vs. layer count N , vs. particles per layer N_p , vs. loaded scene splat count; breakdown sim / sort / raster / post. The headless harness already exposes the hooks. Replace the qualitative

claims above with these numbers.] For calibration of the design point (not of solver efficiency—these systems integrate far larger simulation state), the closest prior systems report 2.37 s per frame (FieryGS (Shen et al. 2026), RTX 4090D) and 38–228 s per *time step* (Gaussian Fluids (Xing et al. 2025), RTX 4090); production layered fire required ~ 20 s per slice per frame at $\sim 2048^2$ slice resolution on a GPU farm (Horvath and Geiger 2009), versus our 32×48 —much of the gap is resolution deliberately traded away, the rest is architecture.



Figure 3. Source hot-swap through the layer contract, same renderer and parameters. Left: all nine layers driven by the kinematic curl-noise plume—adequate bulk shape, but uniform ascent without vortical rolling. Right: all layers driven by the vortex-in-cell solver—baroclinic vorticity rolls the plume edges and produces the characteristic unsteady flame silhouette. Layers can be swapped per-layer at runtime with zero renderer changes.



Figure 4. Per-primitive occlusion with a loaded 3DGS scene (synthetic test reconstruction: ground, back wall, and two pillars—one in front of the flame, one behind). Left: bucketed depth interleaving (§5.5): the front pillar correctly occludes the flame, the flame occludes the rear pillar and wall. Right: naive draw-after compositing (prior behavior, and the analogue of backdrop compositing (Shen et al. 2026)): the flame renders in front of everything, and both pillars vanish entirely—without an intra-scene sort, the wall and ground drawn earlier exhaust the alpha budget of pixels that physically belong to the pillars. Same simulation state in both images.



Figure 5. Slice orientation, viewed from the side (90° orbit). Left: view-aligned slices re-orient with the camera; the volume reads correctly from any azimuth (peak brightness 758 / 758 side/front). Right: world-fixed slices seen edge-on collapse into stripes (peak brightness drops to 531)—the failure mode that motivates view alignment.



Figure 6. Inter-layer coupling (Eq. 3). Left: $c=0$, independent slices—the stack reads as several superimposed fires with uncorrelated sway. Right: $c=1$, slices slaved to the coarse solver—one coherent but overly smooth column. The default $c=0.5$ (Figure 1) preserves per-slice turbulence on a shared bulk motion. [TODO: add the quantitative coherence-vs- c curve to complement the inter-slice-coupling probe reported in the text.]

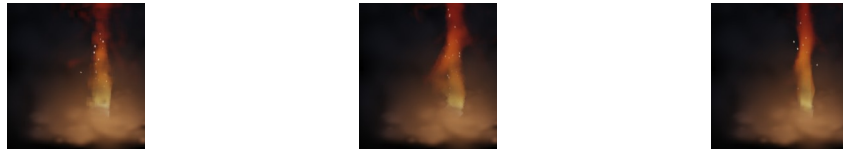


Figure 7. Covariance dynamics. Left: isotropic splats ($k_z=0$) read as soft blobs. Middle: flow-driven covariance (Eq. 5) shears splats into flame licks. Right: adding the trajectory filament stretch (Eq. 6) elongates fast particles into path-aligned streaks.



Figure 8. Layered 2.5D vs. a matched-budget kinematic 3D mode. Left: nine coupled 2D VIC slices (default). Right: the same total particle budget advected by analytic 3D curl noise with entrainment—a volumetric alternative that lacks the baroclinic rolling produced by the slice solvers. [TODO: replace right with a genuine 3D grid/vortex reference solve for the honest comparison promised in the text, and add a perceptual metric.]



Figure 9. HDR post ablation: without (left) and with (right) bloom before ACES tone mapping. Bloom carries the perceived incandescence of the emission-dominated core.

Slice orientation (quantitative).

Using the headless harness we orbit the camera to 0° and 90° azimuth and record the same peak-brightness probe as §4.3 (maximum per-pixel summed RGB, range 0–765). View-aligned slices are viewpoint-isotropic (758 vs. 758); world-fixed slices lose 30 % of peak brightness edge-on (531) and visually collapse into stripes (Figure 5). This quantifies the compositional argument for view alignment inherited from Horvath and Geiger (2009), at zero re-simulation cost in our

Lagrangian setting. **[TODO: sweep azimuth densely (e.g. every 10°) and plot the full curve; add a silhouette-coverage metric alongside peak brightness.]**

Inter-slice coupling (quantitative).

To verify that the operator of Eq. 4 does what it is designed to do—correlate neighboring slices’ turbulent detail—we compute the mean Pearson correlation of the per-slice velocity fields between adjacent slices, averaged over the stack. Isolating the operator with the global coupling off ($c=0$), the probe rises monotonically with operator strength: 0.21 at $\kappa_z=0$, 0.39 at $\kappa_z=0.3$, and 0.61 at $\kappa_z=0.6$. The two coupling mechanisms are thus separable and act on different scales—the global nudge (Eq. 3) aligns bulk motion, the inter-slice term (Eq. 4) aligns local neighbors. We stress this is a coherence probe, not a fidelity measure: it confirms the mechanism, but whether the coupled stack is statistically *closer to 3D* requires a turbulent-energy-spectrum comparison against a true-3D reference solve, which we leave to future work (§8). **[TODO: run the $E(k)$ / cross-correlation comparison against a genuine 3D grid-or-vortex reference; the current 3D mode is kinematic curl noise, not a reference solve.]**

Scene occlusion.

Figure 4 exercises the compositing claim directly: a loaded 3DGS scene with a pillar between the camera and the flame. With bucketed depth interleaving the pillar occludes the flame per primitive; with the naive draw-after pass (equivalent in spirit to backdrop compositing (Shen et al. 2026)) the flame incorrectly renders in front of the entire scene—and the unsorted scene additionally destroys its own internal ordering. The interleave is free: 3.5 vs. 3.6 ms per frame at 3.2k scene splats, 13.0 vs. 13.3 ms at 79k (both within a 60 FPS budget; informal measurements on our development laptop).

Ablations.

Figures 3–9 ablate the major components: simulated vs. procedural sources, coupling strength, covariance dynamics, the layered representation itself, and HDR post. Each toggle is live in the interactive prototype. **[TODO: Add the two ablations that define “layered” : a sweep over slice count $N \in \{1, 3, 5, 9, 17\}$ at fixed total particle budget (visual quality + isotropy probe per N , justifying the default $N=9$), and a sentence justifying stack depth D against the source width. Add a smoke-only scenario (sub-ignition source temperature, longer lifetimes, absorption-dominated shading) so smoke is evaluated as a phenomenon rather than a re-shading preset.]**

Interactive control.

Because dynamics are forward-simulated, the system responds immediately to user input: dragging the fire source drags the plume with correct vortical lag; wind tilts the column; the coupling, buoyancy, baroclinic, and cooling gains are all live sliders; and fire and smoke are two shading presets over the same particle state, switchable per keystroke. **[TODO: capture an interaction sequence for the paper video: source drag, wind, coupling sweep $c: 0 \rightarrow 1$, fire $\leftrightarrow$ smoke toggle, orbit around a loaded .ply scene.]**

Comparisons.

[TODO: Side-by-side against (a) a true 3D grid solver at matched wall-clock budget, (b) an $O(N_p^2)$ Biot–Savart implementation at matched particle count (already in the codebase), and (c) FieryGS on a shared scene if code becomes available. Optional: small user study on perceived realism (FieryGS’ s A/B protocol is a template).]

8 Limitations and Future Work

Physics of stacked 2D slices.

Vortex stretching $(\omega \cdot \nabla)u$ vanishes in 2D, and 2D turbulence cascades energy toward *large* scales, opposite to 3D; a slice stack therefore cannot reproduce the statistics of three-dimensional fire turbulence, however plausible it looks. Our inter-slice coupling operator (Eq. 4) reintroduces the dropped out-of-plane *transport* and measurably raises cross-slice coherence (§7), but it is a diffusive exchange—it does not supply the vortex-stretching *amplification* that drives the forward 3D cascade, so it narrows the gap without closing it. We deliberately trade physical

fidelity for interactivity in a regime—turbulent, shapeless, emission-dominated media—where the eye is forgiving. The honest test is a turbulent-energy-spectrum $E(k)$ comparison against a true-3D reference solve (planned; §7); until it is run, we claim the operator improves inter-slice coherence, not 3D fidelity.

Illumination.

Fire-to-scene lighting is a screen-space approximation, not transport: Gaussian scene splats are warmed by proximity to the flame’s image-space position. Correct volumetric emission from Gaussian mixtures is available in a ray-traced formulation (Condor et al. 2025); a hybrid that shades nearby scene splats from the flame’s particle set (a few thousand point lights, amenable to clustering) is future work, as are cast shadows and smoke self-shadowing.

Renderer completeness.

Our rasterizer implements the forward 3DGS math (sort, footprint, emission-absorption blend) but not the differentiable backward pass—our demo has no reconstruction objective—nor view-dependent spherical-harmonic color, which isotropic incandescence does not need. The FBM modulation of the splat footprint also departs from the pure Gaussian kernel, so fire splats exported to a stock 3DGS viewer would render softer than shown here; an export-compatibility experiment would sharpen the GS-native claim. The exact CPU sort of the dynamic set is adequate at 10^4 Gaussians and the $O(n)$ scene bucketing scales to 10^5 (§5.5), but ordering between the streams is bucket-granular, and tile-binned GPU sorting remains the right implementation for dense fire inside full reconstructed scenes.

View dependence and multi-view consistency.

View-aligned slices make the fire’s world-space geometry a function of the rendering camera: the same particle lifts to different world positions as the camera orbits, so two simultaneous viewers—or the two eyes of a stereo pair—observe slightly different fires. This is the sharpest tension in the “GS-native” framing: the fire’s *attributes* are camera-independent, but its slice lift is view-adapted, unlike a static reconstruction. The world-fixed mode removes the dependence at the cost of edge-on collapse (Figure 5); for stereo, sharing one lift per eye pair is a cheap fix. More fundamentally, because particles are world-space objects, slices can be re-binned to a new axis without restarting the simulation—a remedy unavailable to raster-state slice methods, which we have not yet implemented. Very fast camera motion likewise changes the decomposition axis mid-shot, and flows predominantly along the view axis remain the weakest case (Horvath and Geiger 2009). **[TODO: quantify the world-space displacement of the emissive volume as a function of azimuth change; demonstrate re-binning.]**

Radiometric level of detail.

In the optically-thin, emission-dominated regime the front-to-back accumulation of Eq. 7 is approximately additive, so distant fire clusters could be merged into fewer Gaussians while conserving the summed emission $c \propto$ and its emission-weighted second moment—a radiance-conserving LOD criterion, distinct from the photometric-loss LOD of opaque-splat hierarchies (Kerbl et al. 2024). This is the concrete route to dense fire inside large reconstructions; we have not implemented it.

Geometry-aware flow.

The loaded scene could also act on the *simulation*, not only the compositing: rasterizing a signed-distance field sampled from the 3DGS scene into each slice’s streamfunction Poisson solve as a free-slip (constant- ψ) boundary would deflect the flame around the captured geometry in real time—the per-slice, Lagrangian counterpart of the occupancy-grid obstacle boundaries that offline scene-combustion imposes (Shen et al. 2026).

Toward reconstruction.

The mapping of Table 2 is differentiable end-to-end in principle. Because the forward model exposes only a handful of interpretable controls (source, wind, buoyancy, baroclinic and coupling gains, cooling), fitting *those* to a video summary statistic—a far lower-dimensional, better-constrained inverse than fitting a free field or free Gaussians (Nazarenius et al. 2025; Yu et al. 2023), and solvable even with gradient-free system identification (Jatavallabhula et al. 2021)—would recover not a frozen radiance field but a *running, editable* simulation. Demonstrating a single fitted example is the natural next step.

9 Conclusion

We have presented a system in which fire and smoke live natively inside Gaussian-splatting scenes: not reconstructed, not ray-marched over a backdrop, but forward-simulated in real time by vortex particles that *are* the rendered Gaussians, and demonstrated it in an interactive prototype with qualitative and initial quantitative ablations. The enabling observation—that Lagrangian vortex elements and 3D Gaussians are the same class of point primitive—collapses the simulation-to-rendering pipeline to an attribute mapping, and a pressure-free two-level vortex-in-cell solver, with an inter-slice coupling operator that has no precedent in layered fire, makes a production-proven architecture interactive. Having demonstrated the pattern across three instances—flame, smoke, and embers—we conjecture that the same particle-set unification extends to other optically thin, surfaceless Lagrangian phenomena (sparks, spray, dust, aurorae), and that GS-native effects are a necessary ingredient if reconstructed scenes are to become fully dynamic worlds.

A Per-Slice Solver Pseudocode

```
step_slice(layer L, dt):
  # 1. temperature (grid, semi-Lagrangian)
  for cell g:
    T'[g] = bilerp(T, x_g - dt*u_prev(x_g))
  heat source band
  cool T' *= (1 - 0.5*kc*dt); clamp
  # 2. baroclinic injection (particles)
  for particle i:
    dTdx = (T'(x_i+h/2) - T'(x_i-h/2)) / h
    G_i = (1-lam*dt)*G_i + dt*beta*dTdx
  # 3. P2G: splat G_i -> omega (bilinear)
  # 4. Poisson: lap(psi) = -omega,
```

```

#    K Jacobi sweeps, psi = 0 on boundary
# 5. u = (dpsi/dy, -dpsi/dx) (centered)
# 6. G2P + advect
for particle i:
    v = bilerp(u, x_i)
    v.y += beta_b * T_i      # buoyancy
    v += c*(U_coarse(x_i) - v) # nudge
    v += turb*curl_noise(x_i,t)
    x_i += dt * v
    T_i -= dt * kappa * T_i^2 # cooling
    d_i *= (1 - 0.3*dt)
    recycle if cold/exhausted/out of bounds

```

B Supplemental Material

[TODO: Video: (1) real-time capture of default fire with orbiting camera; (2) interaction reel (source drag, wind, coupling sweep, fire/smoke toggle); (3) ablation cuts matching Figures 4–8; (4) composition with a reconstructed .ply scene. The interactive HTML prototype itself will be released as supplemental code.]

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